Vectors Quiz, Version B

1.

Restate 310° in two other ways.

310° is between 270° and 360°: below east 360° - 310° = 50°

Compass = **50° south of east**

310° - 360° = -50°

negative = **-50°**

same direction: -50° = 40° east of south = 50° south of east

Grading (2): 1 point for each correct form

2.

Restate 25° south of west in two other ways.

south of west is down and to the left: 180° + 25° = 205°

0-360 = **205°**

205° - 360° = -155°

negative = **-155°**

same direction: 205° = -155° = 65° west of south

Grading (2): 1 point for each correct form

3.

Restate −200° in two other ways.

-200° + 360° = 160°

0-360 = **160°**

160° is 20° short of 180°: up and to the left

compass = **20° north of west**

same direction: 160° = 20° north of west = 70° west of north

Grading (2): 1 point for each correct form

4.

Rey's speeder moves east at 52 m/s while a crosswind pushes it south at 36 m/s. Find the speeder's resultant velocity: magnitude and direction.

$v_x = +52$ m/s $v_y = -36$ m/s $v = \sqrt{((52)^2 + (36)^2)} = \sqrt{4000} = 63$ m/s

$v = \boxed{63 \text{ m/s}}$

$\tan^{-1}\frac{36}{52} = 34.7^\circ$ south of east reference angle

~~$\theta = 34.7^\circ$~~ that points north of east; the wind pushes south

$\theta = 360^\circ - 34.7^\circ = 325.3^\circ$

$\theta = \boxed{325^\circ}$

same direction: $-35^\circ = 35^\circ$ south of east $= 55^\circ$ east of south

Grading (5): sketch 1 • v setup 1 • v and unit 1 • tan setup 1 • direction 1

5.

Dumbo flaps at 24 m/s toward 25° north of west while a gust blowing toward 40° east of north adds 9.0 m/s. Find Dumbo's resultant velocity: magnitude and direction.

Convert to the 0-360° scale

D: 25° north of west $= 180^\circ - 25^\circ = 155^\circ$ G: 40° east of north $= 90^\circ - 40^\circ = 50^\circ$

$D_x = 24 \cos 155^\circ = -21.75$ m/s $D_y = 24 \sin 155^\circ = 10.14$ m/s

$G_x = 9.0 \cos 50^\circ = 5.78$ m/s $G_y = 9.0 \sin 50^\circ = 6.89$ m/s

$R_x = -15.97$ m/s $R_y = 17.04$ m/s

$R = \sqrt{((15.97)^2 + (17.04)^2)} = 23.35$ m/s

$R = \boxed{23 \text{ m/s}}$

$\tan^{-1}\frac{17.04}{15.97} = 46.9^\circ$ reference angle

~~$\theta = 46.9^\circ$~~ the calculator's answer; $R_x < 0$ puts it in quadrant II

$\theta = 180^\circ - 46.9^\circ = 133.1^\circ$

$\theta = \boxed{133^\circ}$

same direction: $-227^\circ = 43^\circ$ west of north $= 47^\circ$ north of west

Grading (5): components 2 • add components 1 • R 1 • direction 1